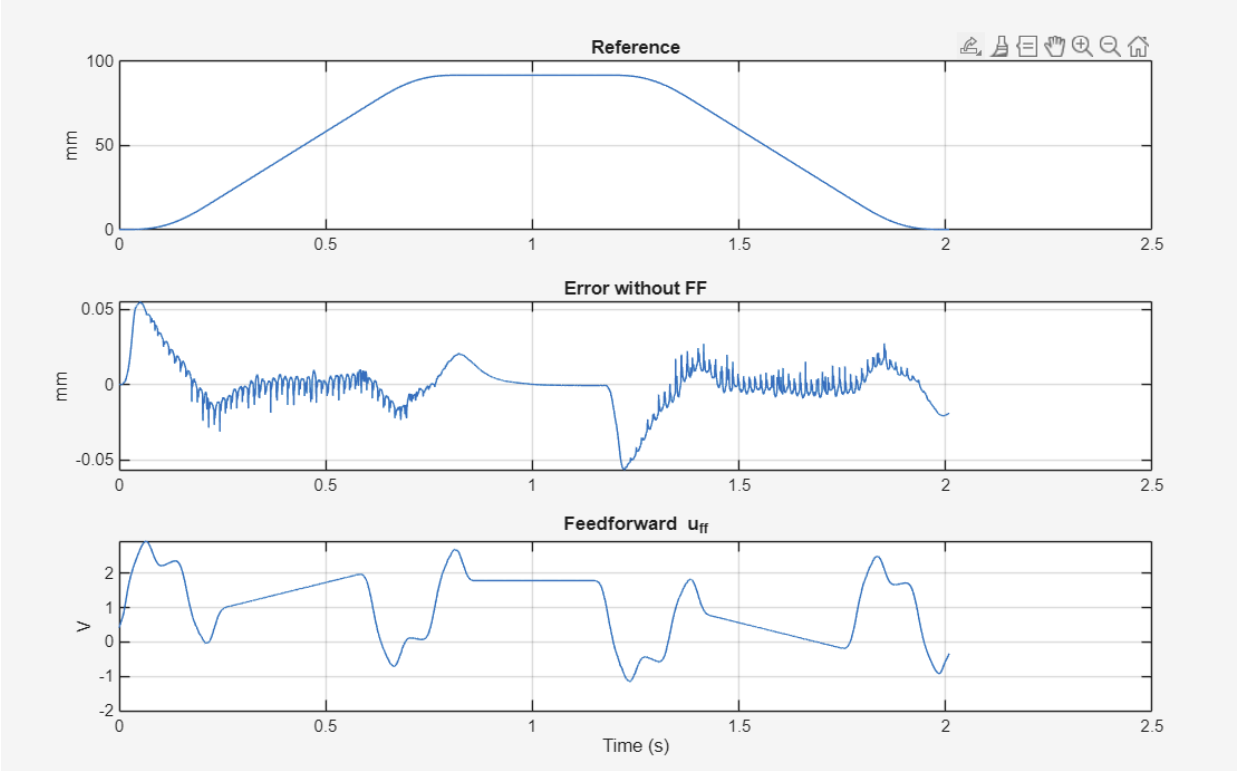
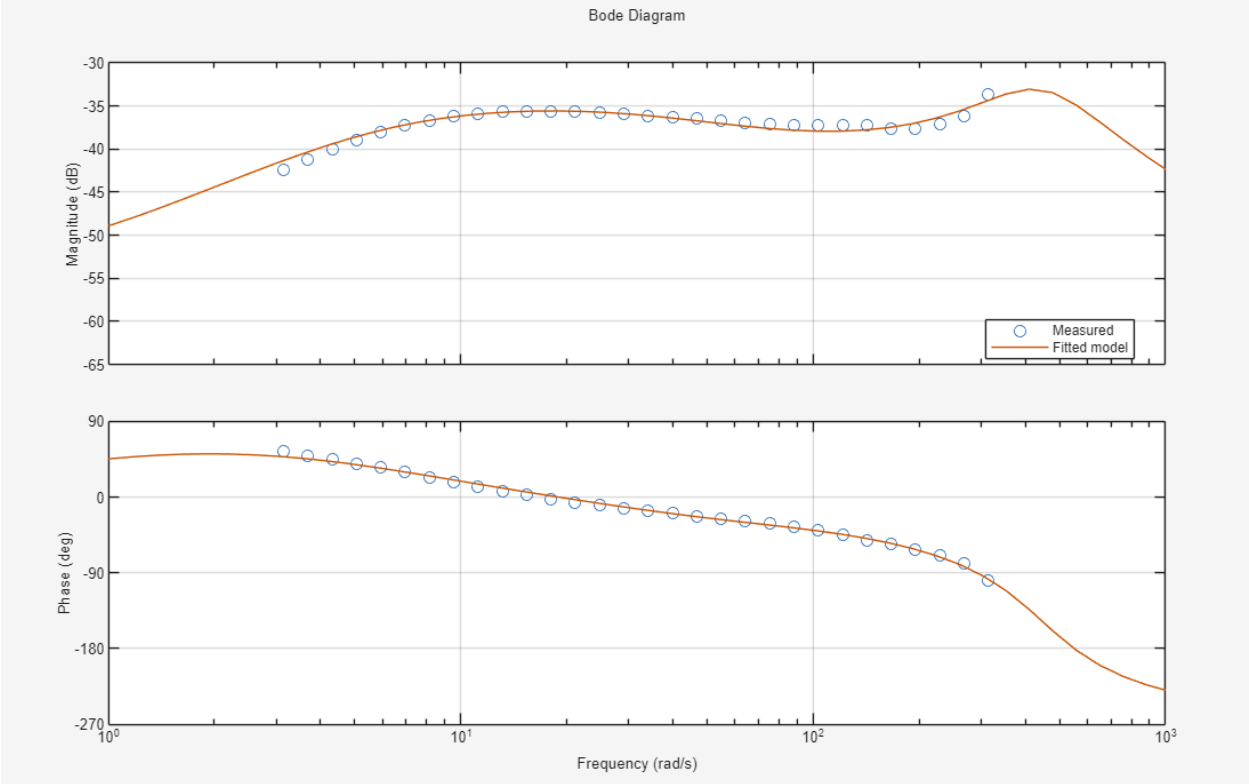
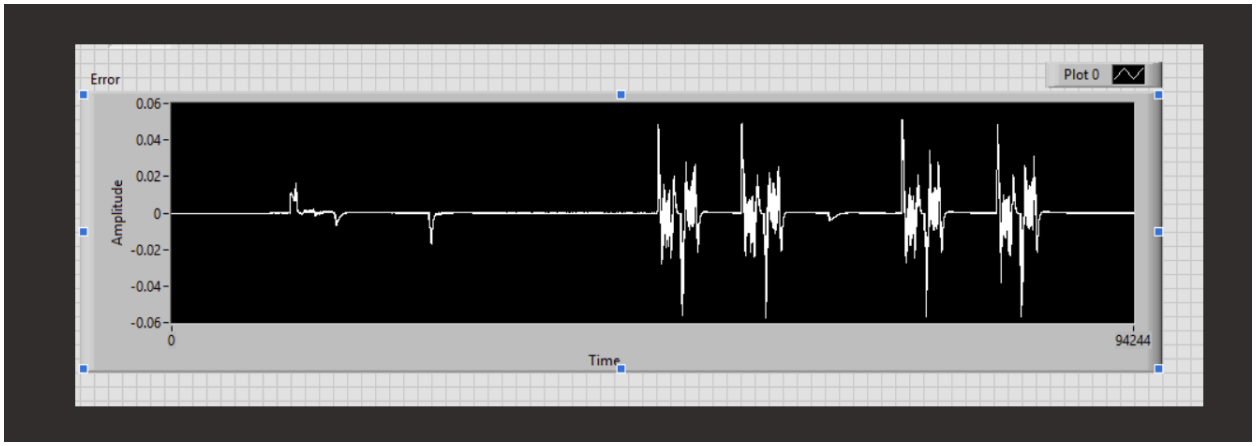


[1] Top: 90 mm 4th order reference. Middle: measured tracking error with feedback only, peaking near 55 µm at motion start. Bottom: feedforward identified by one-shot IFT from that error.



[2] The fitted model matches the measured stage response from 0.5 to 50 Hz.



[3] (error in mm) Tracking error with the lab's nonlinear feedforward model (moves 1 and 2, peak about 10 μm), my one-shot IFT linear feedforward (moves 3 and 4), and feedback only (moves 5 and 6, both peak about 55 μm). Linear feedforward can't remove the friction-driven spikes at motion start; the nonlinear model can.

Controller	RMS tracking error (μm)
Feedback only	50 (measured)
Feedback + my IFT feedforward	7.0 (predicted)
Feedback + published nonlinear feedforward	4.6 (Zhan and Zhou, 2027)[1]

[4] RMS tracking error on the 90 mm 4th order training trajectory. My one-shot IFT linear feedforward was predicted to cut the feedback-only error to 7.0 μm. On the stage, I applied half the feedforward as a cautious first step, and the measured error only dropped slightly. The remaining error is dominated by friction and vibration at motion start, which a linear feedforward can't cancel. The published nonlinear model, which adds friction and cable hysteresis terms, reaches 4.6 μm on the same trajectory. This shows the nonlinear terms matter on this stage, and my current project designs a training trajectory that excites those dynamics so they can be identified better.

[1] S. Zhan and L. Zhou, "A one-shot nonlinear feedforward design approach using ILC-derived signals for precision motion control," *Precision Engineering*, vol. 103, pp. 248-260, 2027. https://doi.org/10.1016/j.precisioneng.2026.08.021